

NATALIA BORREGO, PhD

Behavioral Ecologist · Wildlife Ecologist

Max Planck Institute of Animal Behavior · University of Konstanz · University of Minnesota Lion Center
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RESEARCH AREAS

behavioral ecology · animal behavior · collective behavior · evolutionary ecology · wildlife ecology · sensory ecology

ACADEMIC & RESEARCH APPOINTMENTS

2025–present	ZENiT Research Fellow Zukunftskolleg, University of Konstanz
2022–present	Project Leader Centre for the Advanced Study of Collective Behavior, University of Konstanz
2021–present	Co-Principal Investigator & Postdoctoral Researcher Max Planck Institute of Animal Behavior, Department for the Ecology of Animal Societies
2016–present	Research Associate Lion Research Center, University of Minnesota
2018–2021	Postdoctoral Teaching Fellow at the rank of Assistant Professor Department of Biology, American University in Cairo
2017–2019	Affiliated Faculty Animal Behavior and Conservation Program, Hunter College, CUNY
2017–2018	Visiting Assistant Professor Department of Natural Sciences, Baruch College, CUNY
2017–2018	Research Associate School of Life Sciences, University of KwaZulu-Natal
2016–2017	Postdoctoral Scholar School of Life Sciences, University of KwaZulu-Natal

EDUCATION

2016	PhD, Biology · University of Miami Dissertation: <i>Cognitive complexity in Carnivora: The evolutionary link between social complexity and cognitive complexity</i>
2011	MSc, Environmental Science · Florida Gulf Coast University Thesis: <i>Mercury accumulation in South Florida Blackbirds</i>
2005	BSc, Wildlife Ecology and Conservation · University of Florida

RESEARCH METHODS & FIELD EXPERTISE

Field & behavioral methods: direct observation · experimental behavior & cognition · Indigenous tracking · wildlife capture collaboration · long-term field monitoring · whole-group biologging

Biologging & sensing: GPS telemetry · accelerometry · magnetometry · collar-mounted acoustic monitoring ·

Quantitative analysis & software: parametric and non-parametric statistics · generalized linear and mixed-effects models · QGIS · R · JMP · SPSS · Audacity · Kaleidoscope

RESEARCH FUNDING & FELLOWSHIPS

2025	ZENit Award · Zukunftskolleg, University of Konstanz · Co-PI	€87,000
2022	CASCB Newcomer Grant · University of Konstanz · Co-PI	€305,978
2021	Collaborative Research Grant · MPI-AB / University of Konstanz · Co-PI	€146,728 + salary
2022	CASCB Small Grant · PI	€9,500
2019–2021	American University in Cairo Research Support Grant · PI	\$6,861
2018–2020	National Geographic Council for Exploration Grant · team member	\$27,000
2016	Animal Behavior Society Walter Clyde Allee Honorable Mention	\$750
2016–2017	University of KwaZulu-Natal Postdoctoral Scholarship	\$27,692
2013	University of Miami Dean's Summer Research Fellowship	\$5,000
2012–2014	University of Miami Kushlan Fund	\$4,100
2011–2013	NSF Science Made Sensible Fellowship	\$60,000
2009–2015	Florida Education Fund McKnight Doctoral Fellowship	\$51,000
2009–2011	NIH Initiative for Maximizing Student Development Fellowship	\$30,900
2010	University of Miami Jay M. Savage Award in Tropical Biology	\$400

RESEARCH LEADERSHIP & SCIENTIFIC COLLABORATIONS

Scientific Collaborations

Genevieve E. Finerty MPI-AB / University of Konstanz	African lion behavioral ecology · collective behavior · biologging · comparative field ecology
Margaret C. Crofoot MPI-AB / University of Konstanz	Collective behavior · social dynamics · biologging · comparative behavioral ecology
Craig Packer University of Minnesota Lion Center	African lion social ecology · demography · long-term population research
Simon Chamailé-Jammes CEFE / CNRS, France	Behavioral and sensory ecology · biologging · quantitative ecology
Matthew Wijers WildCRU, University of Oxford	African lion behavior · sensory ecology · multisensor biologging · machine learning
Paul-Antoine Libourel CRNL / CNRS, France	Neurobiology · sensory processing · field neurophysiology

SCIENTIFIC COLLABORATIONS & RESEARCH PARTNERSHIPS

Field, Management & Research Partnerships

Leopard Ecology & Conservation · Botswana	Long-term lion field research · hunting ecology · Indigenous tracking · field-team training
CyberTracker	Tracking methodology · professional certification support · collaborative training for field partners
Ombonde Research Team / Lion Rangers · Namibia	Desert-adapted lion ecology · long-term monitoring · field research · ranger capacity building
Conservancy · Zimbabwe	Whole-pride biologging · behavioral monitoring · comparative lion research
SANParks Scientific Services · South Africa	African lion ecology · management · applied research collaboration
Lion Management Forum of South Africa (LiMF)	Scientific exchange · African lion ecology · evidence-based management

PUBLICATIONS

* equal contribution / co-first author · † joint corresponding author · ‡ student or mentee coauthor

Peer-Reviewed Journal Articles

- 2026** **CO-FIRST AUTHOR · EQUAL CONTRIBUTION**
Chakrabarti, S.*†, Finerty, G. E.*†, **Borrego, N.*†**, Marella, V. N. S. N.‡, Bump, J. K., & Crofoot, M. C. Beyond the Serengeti lion: addressing ecological and systemic biases in behavioural research through horizontal comparison. *Animal Behaviour*, 235, 123562. doi:10.1016/j.anbehav.2026.123562
- 2026** Heydinger, J., Muzuma, U., Hoth-Hanssen, T., Finerty, G. E., **Borrego, N.**, & Beasley, J. C. Bottom-up space use with top-down temporal risk buffering in arid herbivore communities. *Ecology and Evolution*, 16(1), e72836.
- 2024** **CO-FIRST AUTHOR · EQUAL CONTRIBUTION**
Finerty, G. E.*†, **Borrego, N.*†**, Alibhai, S. K., Jewell, Z. C., Tschanz, P., Balone, T.‡, Gabaikanye, T., Gana, M., Monnaanoka, S., Mamou, M., Pudidaroma, S., Tshiama, M., Tshiama, M., Araldi, A., Crofoot, M. C., Henley, S., Ithuteng, P. A., & Schiess-Meier, M. The crossroads of tradition and modern technology: integrative approaches to studying carnivores in low density ecosystems. *Frontiers in Conservation Science*, 5, 1402500. doi:10.3389/fcosc.2024.1402500
- 2024** Conquet, E., Paniw, M., **Borrego, N.**, Nater, C. R., Packer, C., & Ozgul, A. Multifaceted density dependence: social structure and seasonality effects on Serengeti lion demography. *Journal of Animal Ecology*, 93(10), 1493–1509. doi:10.1111/1365-2656.14158
- 2023** Williams, H. J., Sridhar, V. H., Hurme, E., Gall, G. E., **Borrego, N.**, Finerty, G. E., Couzin, I. D., Galizia, C. G., Dominy, N. J., Rowland, H. M., & Hauber, M. E. Sensory collectives in natural systems. *eLife*, 12, e88028. doi:10.7554/eLife.88028
- 2022** O'Connor, V.‡, Chase, S., Chodorow, M., & **Borrego, N.** Exploring innovation and behavioral flexibility in African lions (*Panthera leo*) and snow leopards (*Panthera uncia*). *Behavioural Processes*, 199, 104648. doi:10.1016/j.beproc.2022.104648

PUBLICATIONS

Peer-Reviewed Journal Articles — continued

- 2022** Burkhardt, J. C., Gupta, S., **Borrego, N.**, Heilbronner, S., & Packer, C. Oxytocin increases prosocial behavior in groups of African lions. *iScience*, 25, 104049. doi:10.1016/j.isci.2022.104049
- 2020** **Borrego, N.** Socially tolerant lions (*Panthera leo*) solve a novel cooperative problem. *Animal Cognition*, 23, 327–336. doi:10.1007/s10071-019-01336-4
- 2019** McEvoy, O., Miller, S. M., Beets, W., **Borrego, N.**, Burger, A., Courtenay, B., Ferreira, S., Hanekom, C., Hofmeyr, M., Packer, C., Robertson, D., Stratford, K., Slotow, R., & Parker, D. M. Use of contraceptive techniques as management tools to mimic natural reproductive rates in managed African lions (*Panthera leo*). *Wildlife Research*, 46, 398–408. doi:10.1071/WR18079
- 2018** **Borrego, N.**, Ozgul, A., Slotow, R., & Packer, C. Lion population dynamics: do nomadic males matter? *Behavioral Ecology*, 29, 660–666. doi:10.1093/beheco/ary018
- 2017** **Borrego, N.** Big cats as a model system for the study of the evolution of intelligence. *Behavioural Processes*. doi:10.1016/j.beproc.2017.03.010
- 2016** **Borrego, N.** & Dowling, B. Lions (*Panthera leo*) solve, learn, and remember a novel resource acquisition problem. *Animal Cognition*, 19(5), 1019–1025. doi:10.1007/s10071-016-1009-y
- 2016** **Borrego, N.** & Gaines, M. Social carnivores outperform asocial carnivores on an innovative problem. *Animal Behaviour*, 114, 21–26. doi:10.1016/j.anbehav.2016.01.013
- 2008** **Borrego, N.**, Ozgul, A., Armitage, K. B., Blumstein, D. T., & Oli, M. K. Spatiotemporal variation in survival of male yellow-bellied marmots. *Journal of Mammalogy*, 89, 365–373. doi:10.1644/07-MAMM-A-061R1.1

OTHER SCHOLARLY OUTPUTS

Peer-Reviewed Conference Proceedings

- 2026** **VMV 2026 proceedings paper** — accepted for open-access publication and presentation at VMV 2026, Siegen, Germany, 22–25 September 2026. Full camera-ready citation to be inserted when final title and author list are available.
- 2025** **CO-FIRST AUTHOR · EQUAL CONTRIBUTION**
Borrego, N.*†, Finerty, G. E.*†, Balone, T.‡, Gabaikanye, T., Gana, M., Monnaanoka, S., Mamou, M., Pudidaroma, S., Tshiana, M., Tshiana, M., Araldi, A., Crofoot, M. C., Henley, S., Ithuteng, P., & Schiess-Meier, M. Beyond observation: Indigenous tracking explores complex lion behaviour in challenging environments. *African Journal of Wildlife Research*, 55, 405. doi:10.3957/056.055.0405

Manuscripts Under Review

- 2026** Heydinger, J., Muzuma, U., Finerty, G. E., **Borrego, N.**, Horth-Hanssen, T., & Beasley, J. Prey availability and reproductive demands organize space use across temporal scales in desert-adapted lions. Under review at *Movement Ecology*.
- 2026** Slania, N. E., Demartsev, V., Bousquet, C., **Borrego, N.**, Berthet, M., Strandburg-Peshkin, A., Reina, A., Chaverri, G., Chimento, M., Finerty, G. E., Kohles, J. E., Montgomery, T. M., Razik, I., van der Veer, A. H., & Gall, G. E. C. Proto-gossip: investigating precursors to human gossip in nonhuman animals. Submitted to *Biological Reviews*.

Book Chapters & Public-Facing Scholarly Writing

- 2025** **Borrego, N.** The most charismatic of cats. In *The Last Lions*. Smithsonian Books, pp. 44–55.
- 2023** Palmer, M. S., **Borrego, N.**, & Packer, C. Social strategies of the African lion. In *Social Strategies of Carnivorous Mammalian Predators: Hunting and Surviving as Families*. Springer International Publishing, pp. 7–45.

Technical & Professional Publications

- 2018** Miller, S. M., Ferreira, S., Kilian, H., Parker, D., Courtenay, B., Hanekom, C., & **Borrego, N.** Reintroduction, genetic management and genetic rescue of lion populations. In *IUCN SSC Cat Specialist Group, Guidelines for the Conservation of Lions in Africa*, Version 1.0, p. 111.

INVITED TALKS & CONFERENCE PRESENTATIONS

Invited Talks, Symposia & Panels

- 2026** **Borrego, N.**, Chakrabarti, S., & Finerty, G. E. *Beyond Model Systems: Addressing ecological and systemic biases in behavioural research*. FINE (Frontiers in Social Evolution) seminar series.
- 2025** Chakrabarti, S., **Borrego, N.**, & Finerty, G. E. *Understanding lion societies: The role of comparative research*. Symposium speaker and panelist, Behaviour 2025: XXXVIII International Ethological Congress.
- 2025** **Borrego, N.** *Big Cats, Bigger Questions: Proximate and ultimate drivers of behavior*. Max Planck Institute for Evolutionary Anthropology, Leipzig.
- 2025** **Borrego, N.** & Finerty, G. E. *Dynamic Behaviors, Dynamic Landscapes: Flexibility and adaptation in lion behavior*. Human-Carnivore Co-Adaptation Workshop, University of Cologne, Kasane, Botswana.
- 2021** **Borrego, N.** *Thinking like a lion: an adventure in animal research*. MaxCine Science Goes Public, Max Planck Institute of Animal Behavior.

Contributed Presentations

- 2026** **Borrego, N.** *Why Lions Hunt Alone: New Perspectives on Social Carnivores*. Savannah Science Network Meeting.
- 2025** **Borrego, N.** & Finerty, G. E. *Beyond observation: emerging technologies unlock new frontiers in lion behavioral ecology*. South African Wildlife Management Association.
- 2024** **Borrego, N.** & Finerty, G. E. *The crossroads of tradition and modern technology*. South African Wildlife Management Association.
- 2024** **Borrego, N.** & Finerty, G. E. *Pride & Seek — tracking African lion behavior*. Savannah Science Network Meeting. [Poster]
- 2024** **Borrego, N.** & Finerty, G. E. *The crossroads of tradition and technology*. ASAB Winter Meeting, Edinburgh.
- 2024** **Borrego, N.**, Chakrabarti, S., & Finerty, G. E. *Building bridges from boxes: The role of under-studied populations in expanding the horizons of animal behavior*. ASAB Winter Meeting, Edinburgh.

RESEARCH SUPERVISION & MENTORSHIP

- 2026** **Candice Bazin** · MSc research, Sorbonne Paris Nord University
Hunting in complex landscapes: fine-scale analysis of lion hunting adaptations to vegetation cover using traditional and modern tools
Outcome: graduated first in her cohort with the highest thesis grade.
- 2023–present** **Niharika Marella** · Project intern and guest affiliate, Max Planck Institute of Animal Behavior
Geographic bias in lion field research and comparative behavioral ecology.
Outcome: coauthor, *Animal Behaviour* (2026); now leading independent first-author work.
- 2024–2025** **Katja Regan** · BSc Honors, University of Konstanz
How does habitat diversity shape lion behavior in high-density systems?
- 2024–2025** **Anna Greszki** · BSc Honors, University of Konstanz
Investigating the influence of temporal and spatial abiotic drivers on the behavior of male lions
- 2024–2025** **Luisa Ketterer** · BSc Honors, University of Konstanz
The influence of home range and neighbor dynamics on territorial behavior in lions
- 2019–present** **Samar El Sayad** · MSc, Beni-Suef University, Cairo
Behavioral and biochemical studies of captive big cats in Egypt
- 2017–2019** **Victoria L. O'Connor** · MA, Hunter College, CUNY
Exploring innovation and behavioral flexibility in African lions and snow leopards
Outcome: conference presentations and O'Connor et al. (2022), *Behavioural Processes*; continued to doctoral research, with Borrego later serving on her doctoral advisory committee.

Selected Student & Mentee Research Presentations

- 2026** Balone, T., Adler, K., Araldi, A., **Borrego, N.**, Packer, C., Wjiers, M., & Finerty, G. E. *Foraging behavioral shifts of lions at the interface of reserve and cattle-post landscapes*. Savanna Science Network Meeting.
- 2020** O'Connor, V., **Borrego, N.**, & Vonk, J. *Exploring innovation and behavioral flexibility in jaguars, Amur leopards and clouded leopards*. Annual Meeting of the Animal Behavior Society.
- 2019** O'Connor, V., Chase, S., Thomas, P., & **Borrego, N.** *Exploring problem-solving and innovation in snow leopards*. 56th Annual Meeting of the Animal Behaviour Society and 26th International Conference on Comparative Cognition.

UNIVERSITY TEACHING EXPERIENCE

- 2018–2021** **American University in Cairo** · Postdoctoral Teaching Fellow at the rank of Assistant Professor
- *Scientific Thinking: Animal Behavior*
 - *Scientific Thinking: Evolution*
- 2017–2018** **Baruch College, City University of New York** · Visiting Assistant Professor
- *Fundamentals of Ecological Research* — lecture and laboratory
 - *Conservation Biology and Sustainable Development* — lecture and laboratory
- 2014–2016** **University of Miami** · Teaching Assistant
- *Evolution and Biodiversity Laboratory*
- 2007–2009** **Florida Gulf Coast University** · Adjunct Instructor
- *Biology II Laboratory*

RESEARCH MENTORSHIP & CAPACITY BUILDING

- 2024–2026** **Mentor, African Research Talents Experiencing Mentoring in Science (ARTEMIS) Max Planck Institute of Animal Behavior**
Mentorship of early-career African researchers developing independent projects and research careers.
- 2023–present** **Uakendisa Muzuma** · ARTEMIS mentee
Spatial ecology of the human–lion interface in arid north-western conservancies, Kunene Region, Namibia; hosted for the 2025 Zukunftskolleg ZUKOnnect Fellowship.
- 2025** **Joshua Overmeyer** · ARTEMIS mentee
Variation in leopard density and activity across a multi-use landscape in the Western Cape, South Africa.
- ongoing** **Trevor Balone** · Leopard Ecology & Conservation, Botswana
Long-term field research assistant and mentee in the Kalahari lion research programme.
Outcomes: coauthor on 2024 and 2025 peer-reviewed publications; presented KSI research at the 5th Botswana Biodiversity Symposium; earned professional certifications and gained acceptance into the University of Oxford's postgraduate diploma program.
- 2018–present** **CyberTracker / LEC tracking-team capacity building**
Supported CyberTracker training workshops for the Botswana tracking team, initially through the National Geographic Council for Exploration project and subsequently through the Collaborative Research Grant.
- 2022–present** **Lion Rangers capacity building, Namibia**
Co-hosted SMART training for Lion Rangers through the CASC Newcomer Grant.

PROFESSIONAL & ACADEMIC SERVICE

Graduate Examination & Advisory Service

- 2024** **Doctoral Advisory Committee Member** · Oakland University
Victoria L. O'Connor, DSc Psychology. *Exploring Innovation and Behavioral Flexibility in Captive Carnivores*.
- 2019** **MSc Thesis Examiner** · Beni-Suef University
Shimaa Husien Mohamed Abd-elwhab. *Using of algal free cells, treated and biofilms for industrial wastewater treatment*.
- 2017–2020** **PhD Committee Member** · University of KwaZulu-Natal / Leopard Ecology & Conservation
Kietumetse Ngaka. Thesis topic: lion predation in extreme environments.

Professional Scientific Service

- Ongoing** **Peer Reviewer**
International journals in behavioral ecology, animal behavior, conservation, and wildlife science.
- Current** **Scientific Member** · Lion Management Forum of South Africa (LiMF)
Scientific exchange on African lion ecology and evidence-based management.

SCIENTIFIC LEADERSHIP & COLLABORATIVE WORKING GROUPS

- 2026** **Co-host, Sensory Ecology Workshop** · Zukunftskolleg / University of Konstanz
ZENiT-supported interdisciplinary workshop bringing together collaborators and external expertise to develop approaches for studying sensory mechanisms underlying animal behavior.
- 2026** **Comparative Animal Gossip Working Group**
Participated in interdisciplinary workshop and manuscript development; resulting synthesis manuscript is under review at *Biological Reviews*.
- 2023** **Sensory Collectives Working Group**
Participated in interdisciplinary workshop and synthesis collaboration culminating in Williams et al. (2023), *eLife*.

TEACHING & MENTORSHIP DEVELOPMENT

- 2024–2026** **ARTEMIS mentor development workshops**
Cross-cultural mentorship and research capacity building with graduate and early-career researchers from African institutions.
- 2018–2019** **Teaching Advancement Program**
American University in Cairo Center for Learning and Teaching
Faculty Institute of Learning and Teaching Immersion Track.
- 2018** **Teaching Enhancement Certificate**
American University in Cairo Center for Learning and Teaching

EARLIER EDUCATIONAL OUTREACH & RESEARCH MENTORING

- 2011–2013** **NSF Science Made Sensible Fellow / Resident Scientist**
Allapattah Middle School and Carol City Middle School, Florida; science engagement and research-based instruction.
- 2013** **Graduate Mentor, Research in Ecology Program** · University of Miami / Howard Hughes Medical Institute
Primate Cognition and Behavior Research Group.

SELECTED PUBLIC SCHOLARSHIP, SCIENCE COMMUNICATION & MEDIA

Featured Scientist & Broadcast

- 2026** **Max Planck Forschung 3/2026 — “Natalia Borrego”**. Six-page *Besuch bei* featured-scientist profile by Nora Lessing.
- 2024** **Animal Minds: What Are They Thinking?** — research featured / featured scientist, Orca children’s book.
- 2021** **Animal IQ** — presenter and co-host, eight-episode web series, PBS Nature / PBS TerraNova.
- 2019** **Wicked Weasels** — TV presenter and scientific consultant, BBC / PBS.
- 2018** **Big Cats** — TV presenter and scientific consultant, BBC / PBS.
- 2015** **Killer IQ: Predator Challenge** and **Killer IQ: Predator Games** — TV presenter and scientific consultant, Smithsonian Channel.

Selected Research Coverage & Scientific Consulting

- 2022** **Smithsonian Magazine** — research coverage on oxytocin and lion social behavior.
- 2021** **BBC, Chris Packham’s Animal Einsteins**, Episode 4 “Social Networkers” — research featured.
- 2020** **Earth at Night** — scientific consultant for African lion episode, Apple TV+ / Offspring Films.
- 2018** **National Geographic Online News** — scientific consultant, “Do Big Cats Get Hairballs?”
- 2016** **Scientific American** — press coverage, “Lions are the brainiest of the Big Cats.”
- 2016** **Science** — press coverage, “Good social skills make animals smarter.”
- 2021** **Wildscreen Network Festival** — panelist, “Understanding Science Communication when Filming/Photographing Wildlife.”